

Unit 9: Second-Order Linear Equations

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb through the three characteristic-root cases — real distinct, repeated, and complex conjugate — then exercise the Wronskian, Abel's identity, reduction of order, the Cauchy–Euler equation, and a higher-order capstone. Solutions follow in Part III.

1. **(Real distinct roots)** Solve $y'' - 5y' + 6y = 0$. Write the characteristic equation, factor it, and give the general solution.
2. **(Repeated root)** Solve $y'' - 6y' + 9y = 0$. Identify the double root and explain why the factor x appears in the second solution.
3. **(Complex conjugate roots)** Solve $y'' + 4y' + 13y = 0$. Report the complex roots $\alpha \pm \beta i$ and the real general solution.
4. **(Real-root IVP)** Solve the initial value problem $y'' - y' - 2y = 0$, $y(0) = 1$, $y'(0) = 0$.
5. **(Wronskian)** Compute the Wronskian of $y_1 = e^{2x}$ and $y_2 = e^{3x}$, and state what it certifies.
6. **(Abel's identity)** For $y'' + 3y' + 2y = 0$, use Abel's formula to find the Wronskian up to a constant, then verify it directly from the fundamental solutions.
7. **(Reduction of order)** The equation $x^2y'' - 3xy' + 4y = 0$ has the solution $y_1 = x^2$. Use reduction of order to find a second, independent solution y_2 .
8. **(Cauchy–Euler)** Solve $x^2y'' + 2xy' - 6y = 0$ for $x > 0$.
9. **(Repeated-root IVP)** Solve $y'' - 4y' + 4y = 0$, $y(0) = 1$, $y'(0) = 1$.
10. **(Capstone, higher order)** Solve $y''' - 3y'' + 3y' - y = 0$. Identify the multiplicity of the characteristic root and write the general solution.

Part III — Complete Solutions

Solution 1. Try $y = e^{rx}$ to get the characteristic equation, then factor:

$$r^2 - 5r + 6 = (r - 2)(r - 3) = 0 \Rightarrow r_1 = 2, r_2 = 3.$$

The roots are real and distinct, so

$$y = c_1 e^{2x} + c_2 e^{3x}.$$

Solution 2. The characteristic equation is a perfect square:

$$r^2 - 6r + 9 = (r - 3)^2 = 0 \Rightarrow r = 3 \text{ (double)}.$$

A double root supplies only one exponential e^{3x} ; reduction of order furnishes the second solution $x e^{3x}$, so

$$y = (c_1 + c_2 x) e^{3x}.$$

Solution 3. Complete the discriminant:

$$r = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = -2 \pm 3i.$$

With $\alpha = -2$ and $\beta = 3$,

$$y = e^{-2x} (c_1 \cos 3x + c_2 \sin 3x).$$

Solution 4. Characteristic equation $r^2 - r - 2 = (r - 2)(r + 1) = 0$ gives $r = 2, -1$, so $y = c_1 e^{2x} + c_2 e^{-x}$. Apply the initial conditions:

$$y(0) = c_1 + c_2 = 1, \quad y'(0) = 2c_1 - c_2 = 0.$$

Adding the equations, $3c_1 = 1$, so

$$c_1 = \frac{1}{3}, \quad c_2 = \frac{2}{3}, \quad y = \frac{1}{3} e^{2x} + \frac{2}{3} e^{-x}.$$

Solution 5. With $y_1 = e^{2x}$, $y_2 = e^{3x}$:

$$W = y_1 y_2' - y_2 y_1' = e^{2x} \cdot 3e^{3x} - e^{3x} \cdot 2e^{2x} = (3 - 2)e^{5x} = e^{5x}.$$

Since $e^{5x} \neq 0$ for all x , the two solutions are **linearly independent** and $\{e^{2x}, e^{3x}\}$ is a fundamental set.

Solution 6. Here $p(x) = 3$, so Abel's formula gives

$$W(x) = C \exp\left(-\int 3 dx\right) = C e^{-3x}.$$

Check directly: the roots of $r^2 + 3r + 2 = (r + 1)(r + 2) = 0$ are $r = -1, -2$, so $y_1 = e^{-x}$, $y_2 = e^{-2x}$ and

$$W = e^{-x}(-2e^{-2x}) - e^{-2x}(-e^{-x}) = (-2 + 1)e^{-3x} = -e^{-3x},$$

which matches $C e^{-3x}$ with $C = -1$.

Solution 7. Normalize to $y'' - \frac{3}{x}y' + \frac{4}{x^2}y = 0$, so $p(x) = -3/x$ and $e^{-\int p dx} = e^{\int (3/x) dx} = e^{3 \ln x} = x^3$. With $y_1 = x^2$,

$$y_2 = y_1 \int \frac{e^{-\int p dx}}{y_1^2} dx = x^2 \int \frac{x^3}{x^4} dx = x^2 \int \frac{1}{x} dx = x^2 \ln x.$$

The second independent solution is $y_2 = x^2 \ln x$ (the repeated-root Cauchy–Euler form).

Solution 8. Try $y = x^m$ in $x^2 y'' + 2xy' - 6y = 0$. The indicial equation is

$$m(m-1) + 2m - 6 = m^2 + m - 6 = (m+3)(m-2) = 0 \Rightarrow m = 2, -3.$$

The roots are real and distinct, so for $x > 0$

$$y = c_1 x^2 + c_2 x^{-3}.$$

Solution 9. The characteristic equation $r^2 - 4r + 4 = (r-2)^2 = 0$ has the double root $r = 2$, so $y = (c_1 + c_2 x)e^{2x}$. Differentiate: $y' = (c_2 + 2(c_1 + c_2 x))e^{2x}$. Apply the conditions:

$$y(0) = c_1 = 1, \quad y'(0) = c_2 + 2c_1 = 1 \Rightarrow c_2 = -1.$$

Therefore

$$y = (1 - x)e^{2x}.$$

Solution (Capstone). The characteristic equation is a perfect cube:

$$r^3 - 3r^2 + 3r - 1 = (r-1)^3 = 0 \Rightarrow r = 1 \text{ (multiplicity 3)}.$$

A triple root contributes e^x , xe^x , and $x^2 e^x$, so

$$y = (c_1 + c_2 x + c_3 x^2)e^x.$$

Unit 9 Key Takeaway

Every constant-coefficient solution is read straight off the **characteristic roots**: real-distinct roots give $c_1 e^{r_1 x} + c_2 e^{r_2 x}$, a repeated root carries the factor x as $(c_1 + c_2 x)e^{rx}$, and a complex pair gives $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$. The **Wronskian** confirms independence, **Abel's identity** predicts it, and **reduction of order** recovers the missing partner — exactly the $x^2 \ln x$ that the repeated Cauchy–Euler root demands.